

8 *Rāga and Tāla*



Let us recall

Nāda: Musical sound

Śhruti: A sound that is heard clearly and can be recognised as distinct from another

Svara: A note in music. There are seven notes in Indian music—
S R G M P D N

Saptak/Sthāyī: An octave or register in music. Musicians usually perform in Mandra (lower), Madhya (middle) and Tāra (higher) saptak/octaves.



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Svara

Let us revisit some of the basic building blocks of melody in classical music.

- ◇ There are 7 notes (*svaras*) of music.
- ◇ S and P are the invariant notes known as **achala-svara/prakṛiti-svara**.
- ◇ The remaining 5 notes (R G M D N) have higher and lower variants and known as **chala-svaras/vikṛiti-svaras**.
- ◇ So, the total number of notes (*svara-sthānas*) comprise 2 *prakṛiti-svaras* or *achala-svaras* (S and P), and 2 variants of the remaining 5 notes (R G M D N), bringing the total number of note positions to 12.

In a keyboard or a harmonium, you can see the arrangement of notes in an octave, and how the same note positions repeat, across different octaves.

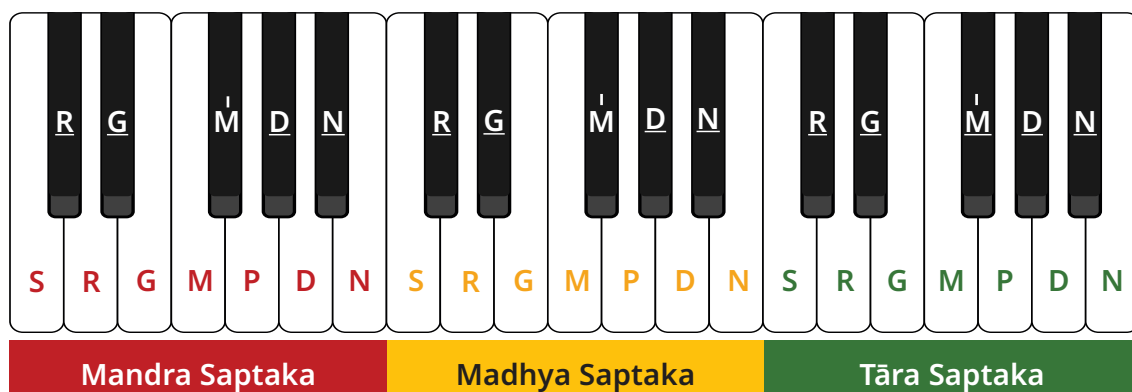


Fig. 8.1. The arrangement of svaras and octaves on a keyboard

It is important to know that the 7 notes and 12 note positions are the same in both Hindustani and Carnatic Music, but the variants are known by different names.



Note to the teacher: If you have access to a keyboard or a harmonium, use it to introduce the concepts of 7 notes, 12 note positions, and octaves. Alternatively, you can also download a piano app on your mobile phone or computer.

Table 8.1. Svara-sthānas in Hindustani and Carnatic Music

Svar	Name	Note Position	Variant Names (Hindustani)	Variant Names (Carnatic)
S	Śhaḍja		No variant	No variant
R	Riṣhabha	Lower	Komala	Śhuddha
		Higher	Śhuddha	Chatuḥśhruti
G	Gāndhāra	Lower	Komala	Sādhāraṇa
		Higher	Śhuddha	Antara
M	Madhyama	Lower	Śhuddha	Śhuddha
		Higher	Tivra	Prati
P	Pañchama		No variant	No variant
D	Dhaivata	Lower	Komala	Śhuddha
		Higher	Śhuddha	Chatuḥśhruti
N	Niṣhāda	Lower	Komala	Kaiśhikī
		Higher	Śhuddha	Kākalī

Do you know?

The 7 notes and 12 note positions are found in most music systems around the world!



Rāga

A *rāga* is a melodic framework in Indian Classical Music and comprises a set of notes. The *rāga* system is one of the reasons why Indian Classical Music is one of the most sophisticated, complex, and scientific music systems in the world.

So, how do we group and classify these hundreds of *rāgas*? Just like you would have studied the classification of plants and animals into different groups and families, *rāgas* are also classified and grouped according to their characteristics so that we can understand them better.

Classification based on number of *svaras*

In both Hindustani and Carnatic traditions, one way of classifying *rāgas* is based on how many notes are used in the *ārohaṇa* (ascending scale) and *avarohaṇa* (descending scale).

In Hindustani Music, this classification of *rāgas* based on the number of notes in the scale is called the Rāga-jāti.



Fig. 8.2. Students performing Carnatic Classical Music (Kala Utsav 2025)

Table 8.2. Rāga-jāti and Examples			
No. of <i>Svaras</i>	Name	Example in Hindustani	Example in Carnatic
7	<i>Sampūrṇa</i>	Alhaiyā Bilāval, Yaman	Śhaṅkarābharaṇam (S R G M P D N S)
6	<i>Śhāḍava</i>	Gurjarī Todī (S R G M D N S)	Śhrīraṅjanī (S R G M D N S)
5	<i>Auḍava</i>	Bhūpālī (S R G P D S)	Haṁsadhvani (S R G P N S)

Rāgas can also have different numbers of notes in the *āroha/ārohaṇa* and *avaroha/avarohaṇa*.

Rāga-jātis

<i>Sam̐pūrṇa — Sam̐pūrṇa</i>	<i>Sam̐pūrṇa — Śhāḍava</i>	<i>Sam̐pūrṇa — Auḍava</i>
<i>Śhāḍava — Sam̐pūrṇa</i>	<i>Śhāḍava — Auḍava</i>	<i>Śhāḍava — Śhāḍava</i>
<i>Auḍava — Sam̐pūrṇa</i>	<i>Auḍava — Auḍava</i>	<i>Auḍava — Śhāḍava</i>

Activity 8.1

Write any *Auḍava Rāga* scale _____

Write any *Śhāḍava Rāga* scale _____

Write any *Auḍava-Śhāḍava* scale _____

Tānsen and the rāga of fire

Legend says that Tānsen, Akbar's favourite singer, was tricked by jealous courtiers into singing *Rāga Dīpaka*, believed to create deadly heat. Knowing only *Rāga Megha* could save him, he trained his daughter *Sarasvatī* and her friend *Rūpavatī*. As the heat rose during his performance, the girls sang *Rāga Megha*, bringing rain and saving Tānsen.



Fig. 8.3. Portrait of Tānsen c. 1585–90, National Museum, New Delhi

Discussion circle

If you could name a *rāga* what would it be? Describe how you would like the *rāga* to make a listener feel. _____

Classification based on the *svara-sthānas*

Rāgas in both Carnatic and Hindustani Music have also been traditionally classified into various families, based on their note positions (*svara-sthānas*). In this case, the systems of classification in Carnatic and Hindustani Music are different. Let us look at each one below.

Carnatic 72 *Meḷakartā* System

The 72 *meḷakartās* form the basic scale system of Carnatic Music. Each *meḷakartā rāga* uses all seven *svaras* in both the *ārohaṇa* (ascending) and *avarohaṇa* (descending) order. In all, 72 different scales are created by systematically combining the lower and higher variants of the *svaras* — R, G, M, D, and N.

Do you know?

The system of mathematically deriving 72 *meḷakartā* scales based on the permutation and combination of *svara-sthānas* was established by Venkatamakhi in his treatise *Chaturdaṇḍī-Prakāśhikā*, which was published in the eighteenth century.

Janaka Janya Rāgas

These 72 *meḷakartās* can also be referred to as *Janaka* (parent) *rāgas*. Many *janya rāgas* are derived from each *janaka* by eliminating notes or changing the order of the notes. For example, the *Meḷakartā/janaka* Rāga Śhaṅkarābharaṇam has the scale:

S	R ₃	G ₃	M ₁	P	D ₃	N ₃	Ś
Ś	N ₃	D ₃	P	M ₁	G ₃	R ₃	S

In Rāga Haṁsadhvani, the M and D *svaras* are removed from Śhaṅkarābharaṇam to give rise to a 5 note (*auḍava*) scale

S	R ₃	G ₃	P	N ₃	Ś
Ś	N ₃	P	G ₃	R ₃	S

Hindustani Thāt System

The *thāt* system is a method used to group *rāgas* based on the notes that are used in the *rāga*. A *thāt* is a musical scale consisting of 7 *svaras* (S R G M P D N) arranged in order. Each of the notes — R G M D, and N have two variants. Based on different combinations of these variants, there are 10 different *thāts*, and all *rāgas* belong to one of these 10 *thāts*.

Eg: The Kalyāṇ Thāt has the scale: S R G M' P D N S

Do you know?

The *thāt* system was developed by musicologist Pandit Vishnu Narayan Bhatkhande in the early twentieth century to make the study of *rāgas* easier and more structured.

Two *rāgas* belonging to the Kalyāṇ Thāt are:

1. *Yaman*

N R G M' D N Ś, Ś N D P M G R S

2. *Śhuddha Kalyāṇ*

S R G P D Ś, Ś N D P M' G R S



Compositions

Let's look at compositions in both Carnatic and Hindustani Music. In both systems, we will deal with a similar *rāga* — Śhaṅkarābharaṇam in Carnatic and Alhaiyā Bilāval in Hindustani.

Carnatic Music — Rāga Śhaṅkarābharaṇam

Feature	Details
<i>Meḷakartā</i> Number	29
Classification (<i>Jāti</i>)	<i>Saṁpūrṇa</i>
Western Equivalent	Major Scale
<i>Ārohaṇam</i> (Ascent)	S R ₂ G ₃ M ₁ P D ₂ N ₃ Ś
<i>Avarohaṇam</i> (Descent)	Ś N ₃ D ₂ P M ₁ G ₃ R ₂ S
Famous Composition	'Enduku Peddala' by Tyāgarāja
Significance	One of the first <i>rāgas</i> students learn due to the simplicity of its scale

Let's learn two compositions that are in *janya rāgas* of Śhaṅkarābharaṇam.

Jatisvaram

It is a Carnatic Music composition that has *svaras* (musical notes) and optionally rhythmic syllables (*jatis*) to create a lively musical piece. It generally has no lyrics or *sāhitya*.

Rāga: Kaṇṇaḍā

Tāla: Ādi

Composer: Veena Padmanābhayya

Ārohaṇa: S G₃ M₁ D₂ N₃ Ś

Avarohaṇa: Ś N₃ Ś D₂ P M₁ G₃ M₁ R₂ S

Pallavi

Ś--- --N- ŚD-P MG-M | D--- --DN | DPM G -M DN ||

Chitte-svara

1. Ś-ŔN -ŚD- D-ND- PM- | G--G -MR- | ṆS-G -MDN ||

2. D--- --DP DP-M GMDP | D--- --DN | ŚD-P MG-M ||

D--D NŚD- NŚ-D NŚŔŚ | D--N DPD- | DPMG -MDN ||

3. Ś--- --ŚŔ N-NŚ D-PM | D--- --DN | D-PD PMGM ||

R--- --SG M-MG MD-D | DNŚŚ -ŔNŚ | DP-M GMDN ||

4. Ś-ŚŔ N-NŚ D-DN DPM- | G-GM R-RG | S-SR SṆS- ||

SG-M DNŚŔ ŚĠ-M ŔŚNR | ŚŚ-D PMGG | -MRS GMDN ||



Svarajati

It is a Carnatic Music composition that combines simple lyrics (*sāhitya*) with musical notes (*svaras*) and optionally, rhythmic patterns (*jati*).

Rāga: *Haṁsadhvani*

Tāla: *Ādi*

Ārohaṇa: S R₂ G₃ P N₃ S

Avarohaṇa: S N₃ P G₃ R₂ S

Pallavi

G <i>Shree</i>	P <i>Raa</i>	G <i>Maa</i>	<u>. R</u> , <i>Bhi</i>	S <i>Raa</i>	<u>. N</u> <i>. Ma</i>	<u>. S</u> , <i>Naa</i>	<u>. R</u> , <i>Du</i>
<u>S</u> <i>Chit</i>	<u>R</u> <i>Tam</i>	<u>. G</u> , <i>Bu</i>	<u>R G</u> <i>Nee Paa</i>	<u>. P</u> , <i>Da</i>	<u>G . N P</u> <i>Pan KaJam</i>	<u>. Ś N P</u> , <i>Bu , Pai</i>	<u>GP . .</u> <i>SaDaa</i> , <u>GRSR</u> <i>YaNiChiTi</i>

Anupallavi

P Nae	. N , Ra	PG MuLa	P Nem	ŚN <u>ChaKu</u>	PN KaRu	PG NaGa	PG Li Gi
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Charaṇa

1.	<u>RG P N</u> NaNuKaRa	<u>Ś R Ś .</u> MuLaChae	<u>Ś NPG</u> BaLiMiGa	<u>. NP .</u> , DaYa ,	<u>G R R .</u> NwiTaPat	<u>S Ś . N</u> Ti Yaa ,Da	P Rin	<u>G R SR</u> ChuMi YiKa
2.	<u>PNŚR</u> NinuViNaa	<u>. GP N</u> , NaNuKru	<u>Ś . R G</u> TaaruThuNi	<u>P N . S</u> GaChae Yu	<u>RGP .</u> TaKuNee	<u>P GRS</u> Ja GaTi Lo	<u>NP GR</u> NeVaRuRa	<u>S N SR</u> NruPaVaRa
3.	<u>GPRG</u> MuNiNuTa	<u>SRSN</u> DaNuJaVi	P Daa	<u>NS R G</u> RaNaSuRa	<u>PNŚR</u> HiTaShuBha	Ś Vee	Ř Ś . N KshaNaa , Bha	Ś N . P YaHa , sTa
4.	<u>P . N .</u> ShreeSee	Ś . R Ś Ta , ViHaa	<u>. N P N</u> , RaChaTu	Ś P Ś . RaSaDaa	<u>P N . G</u> ShiVaarThi	<u>P . RG</u> BhamJaNa	Sr . . SaRo ,	<u>S N PG</u> JaNa YaNa
	Ś R P G MaDaNaJa	Ř Ś R Ś NaKaPaTu	<u>N P GR</u> TaRa Su Va	Ś N P G ChaNaDaSha	Ř Ś R G RaThaSuTa	Ś R N Ś SaDaYaHru	<u>PN GP</u> DaYaViDhi	<u>R GSR</u> HaRaNuTa



Fig. 8.4. Kala Utsav participants playing the sitar and tabla

Hindustani Music — Rāga Alhaiyā Bilāval

Feature	Details
<i>Thāt</i>	Bilāval
<i>Jāti</i>	Sam̐pūrṇa
Notes	In this <i>rāga</i> , all svaras (notes) used are <i>Śhuddha</i> (pure/ natural).
<i>Vādī</i>	D
<i>Sam̐vādī</i>	G
Singing Time	First <i>prahāra</i> * of the night
Nature	This is an Uttaraṅga Pradhāna Rāga
Distinction	Bilāval and Alhaiyā Bilāval are two separate <i>rāgas</i> . Rāga Bilāval is similar to this one, but Komala Niṣhāda is not used in it.
<i>Āroha</i> (Ascending)	S G R G P D N S
<i>Avaroha</i> (Descending)	S N D P, D N D P, M G M R, S
<i>Pakaḍ</i> (Important Phrase)	G R G P D N S

*A day is divided into eight parts. Each part is called a *prahāra* and lasts three hours.

Bandiśh**Tāla: Teen Tāla****Kramik Pustak Mālīka – Bhāṅ 2**

Sthāyī: *Tū Hī Ādhār Sakala Tribhuvana Ko Pālaka Sacharāchar Bhūtan Ko*
Tū Hī Viṣṇu Tū Nārāyaṇa Kāraṇa Tū Parabrahma Jagat Ko

Sthāyī

P			D					N							
G	P	N	N	Ś	-	Ś	Ś	Ś	Ṛ	Ś	N	D	P	<u>MG</u>	<u>MR</u>
Tu	ॐ	Hi	Aa	dha	ॐ	r	Sa	ka	l	Tri	bhu	va	n	<u>koॐ</u>	<u>ॐॐ</u>
3				X				2				0			

M								N							
G	M	P	<u>MG</u>	M	R	S	-	D	N	Ś	N	D	P	<u>MG</u>	<u>MR</u>
Paa	ॐ	la	<u>kॐ</u>	Sa	ch	Raa	ॐ	ch	ar	Bhu	ॐ	Ta	n	<u>koॐ</u>	<u>ॐॐ</u>
3				X				2				0			

Antarā

		P	D					N		Ṛ					
P	-	N	N	Ś	-	Ś	-	Ś	Ḡ	Ḡ	Ṣ	Ḡ	Ṛ	Ś	Ś
Tu	ॐ	Hi	ॐ	Vi	ॐ	shnu	ॐ	Tu	ॐ	Na	ॐ	ra	ॐ	ya	n
3				X				2				0			

N						N		N							
Ś	-	Ḡ	Ṛ	Ś	-	D	P	D	N	Ś	<u>ŚN</u>	D	P	<u>MG</u>	<u>MR</u>
Kaa	ॐ	ra	na	Tu	ॐ	Pa	r	Bram	ॐ	ha	<u>Jaॐ</u>	ga	t	<u>koॐ</u>	<u>ॐॐ</u>
3				X				2				0			



Lakṣhaṇa-Gīta

Tāla: Teen Tāla (Madhya Laya)

Kramik Pustak Mālīka – Bhāg 2

Sthāyī: *Taba Kahata Bilāval Bheda Chatura Jaba Mela Milāvata Śhuddha Surana Ko Prātaḥ Samaya Nit Prathama Prahara Dhaivata Vādī Gā Samvādī Aṣṭa Bheda Saba Gāye Madhura*

Sthāyī

N				G				R				G				D								D							
D		P		M		G		P		—		N		N		Š		—		Š		Š		N		N		P		P	
Ka		ha		t		Bi		La		ɤ		wa		l		Bhe		ɤ		d		Cha		tu		r		Ja		b	
0								3								X								2							

<u>DN</u>	<u>ŠR'</u>	š	<u>ŠN</u>	D	P	M	G	G	R	G	M	G	R	S	—
<u>Meš</u>	<u>šš</u>	l	<u>Miš</u>	laa	š	va	t	Shu	š	dha	Su	ra	n	Ko	š
0				3				X				2			

				D				Ṛ				D			
S	M	G	P	P	—	N	N	Ṣ	Ḡ	Ṣ	Ṣ	N	N	Ṣ	Ṣ
Pra	Ṣ	tah	Sa	may	Ṣ	Ni	t	Pra	tha	m	Pra	ha	r,	Ta	b
0				3				X				2			

Antarā

D								R				G			
P	—	N	N	S	—	S	—	<u>NS</u>	G	G	M	R	—	S	—
Dhai	5	va	t	Va	5	di	5	<u>Ga5</u>	5	Sam	5	va	5	di	5
0		3						X							

M				M											
G	M	P	G	M	R	S	S	<u>DN</u>	<u>SR</u>	S	<u>SN</u>	D	N	S	S
A	5	sht	Bhe	5	d	Sa	b	<u>Ga5</u>	<u>55</u>	ye	<u>Ma5</u>	dhu	r,	Ta	B
0				3				X				2			



Tāla

It is the rhythmic cycle in Indian Classical Music that organises beats into a repeating pattern. Let us revisit the basic *tālas* in Hindustani and Carnatic Music.

Tālas of Hindustani Music

Ṭhekā: The basic pattern of tabla strokes or rhythmic syllables that defines a *tāla* in Hindustani Music.

Mātrā: A single beat within a *tāla*

Vibhāg: A division or group of beats within a *tāla*

Tālī: A stressed beat in a *tāla*, usually marked by a hand clap

Khālī: An unstressed or 'empty' beat in a *tāla*, usually marked by a wave of the hand

Sam: The first and most important beat of a *tāla* cycle



Note to the Teacher: Demonstrate the counting of *tālas* or share a video in class.

Teen Tāla

Mātrā: 16

Tālī: 1, 5, 13 mātrā

Vibhāg: 4

Khālī: 9 mātrā

Mātrā	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Bol	dha	dhin	dhin	dha	dha	dhin	dhin	dha	dha	tin	tin	ta	ta	dhin	dhin	dha
Tālī	X				2				0				3			

Jhaptāla

Mātrā: 10

Tālī: 1, 3, 8 mātrā

Vibhāg: 4 (2, 3, 2, 3)

Khālī: 6 mātrā

Mātrā	1	2	3	4	5	6	7	8	9	10
Bol	dhee	Naa	dhee	dhee	naa	tee	naa	dhee	dhee	naa
Tālī	X		2			0		3		

Activity 8.2

Identifying the various *bols*

- ◈ Listen to a recording of the *ṭheka* being played on the tabla and try and identify its *bols*.



Note to the Teacher: Play a recording of the *ṭheka* of Teen Tāla, or use a mobile app. Assist students as they try and identify the *bols* corresponding to the tabla sounds.

Activity 8.3

Finding *sam*

- ◈ Try and follow along the tabla *ṭheka* of the *bandiśh* being played in class and identify the *sam* as you follow the *tāla*.
- ◈ Observe how the artist, in their elaboration, arrives at the *sam* in different ways.



Note to the Teacher: Play a performance of a *bandiśh* in Teen Tāla or Jhaptāla. Help the students follow the tabla *ṭheka* and identify the *tāla*. Encourage students to appreciate the process by which the artist arrives at the *sam*.

Tālas of Carnatic Music

Carnatic Music also has several *tālas*, of which two most popular are Ādi Tāla and Rūpaka Tāla. Let us revise counting the *tāla* with our fingers as shown below:

Ādi Tāla

Beat	1	2	3	4	5	6	7	8
Syllable	Ta	Ka	Dhi	Mi	Tha	Ka	Ta	Ta
Action	Beat	Little Finger	Ring Finger	Middle Finger	Beat	Turn	Beat	Turn

Rūpaka Tāla

Beat	1	2	3	4	5	6
Syllable	Ta	Ka	Ta	Ka	Dhi	Mi
Action	Beat	Turn	Beat	Little Finger	Ring Finger	Middle Finger

Activity 8.4

Listening to a song and identifying the *tāla* pattern

- ◇ As you listen to the song, first react naturally to the beat by tapping your feet, clapping your hand, etc.
- ◇ Now, try and find how many beats this *tāla* cycle has.



Note to Teacher: Play any song with a strong rhythm in the class.

Activity 8.5

Write a lyric to *tāla*

Tāla is used to arrange *svaras* and lyrics into definite patterns in a song. Fill in the lyrics in any language to fit the below *tāla* cycles with one syllable per beat and then recite it.

Teen Tāla (Hindustani)

Mātrā	1 2 3 4	5 6 7 8	9 10 11 12	13 14 15 16
Example	Once - upon a	Time there lived a	Little girl who	Liked to sing songs
Your lyric				

Rūpaka Tāla (Carnatic)

Mātrā	1 2	3 4 5 6	1 2	3 4 5 6
Example	Ra ma	Ja ya ra ma	Krish-na	Ha re Krish na
Your lyric				

You can use any words in the rhythm, for example, *Tvameva Mātā Pitā Tvameva*.

Activity 8.6

Fitting your song to *tāla*

- ❖ Go back to the song you created in Chapter 6 Activity 6.7 (My family tradition).
- ❖ Does your song naturally fall into a *tāla* or rhythm pattern?
- ❖ How will you modify it (by speeding it up, slowing it down, elongating or compressing a syllable, or adding or removing a word) to fit it to a specific rhythm pattern?
- ❖ Try singing or reciting your song while keeping a *tāla*/steady beat. How does this change the feel of your song?



Through this activity, we observe how *tāla* can provide structure and definition to a song, and indeed this is one of the factors that differentiate prose (which you will find in a story) from song lyrics.

EXERCISES

Q1. Explain the following in one line:

- i. *Ṣhāḍava-sāmpūrṇa*
- ii. *Vibhāg*
- iii. *Jātis* in Hindustani Music

Q2. If a *rāga* has 6 *svaras* in the *ārohaṇa* and 6 in the *avarohaṇa*, what is the *jāti* of the *rāga*. Give an example from the *rāgas* you have learnt.

Q3. Which are *achala-svaras*/ *prakṛiti-svaras* and *chala-svaras*/ *vikṛiti-svaras*? Explain with examples.

Q4. Name two *rāgas* and their scales derived from Śhaṅkarābharaṇam.

Q5. Answer the following:

- i. A *rāga* with 5 notes is called _____.
- ii. The *Meḷakartā* system was devised by _____.
- iii. Vishnu Narayan Bhatkhande grouped the Hindustani *rāgas* in the _____ system.
- iv. Complete the series SMGR , RPMG _____.

Q6. What is the difference between *Jatisvaram* and *Svarajati*?

Q7. Give a brief introduction of Rāga Haṁsadhvani.

Q8. Describe one system of classification of *rāgas* with examples.

Q9. Find out about the contributions of the following stalwarts to Indian music.

- | | |
|--------------|---------------------|
| i. Tānsen | ii. Kumar Gandharva |
| iii. Mīrābāī | iv. MS Subbulakshmi |

Q10. Identify the Rāga-jāti based on the following:

- 5 *svaras* in *āroha* and 6 *svaras* in *avaroha*
- 6 *svaras* in *āroha* and 5 *svaras* in *avaroha*
- 5 *svaras* in *āroha* and *avaroha*
- 7 *svaras* in *āroha* and 7 *svaras* in *avaroha*

Q11. Complete the table.

<i>Svara</i>	Note Position	Variant Names (Hindustani)	Variant Names (Carnatic)
S			
Higher R			
Lower M			
Higher N			
<i>Pañchama</i>			

Q12. Write the *tāla lipi* (notation) of Teen Tāla.

Notes

